

Does the Assistant Check the Record? Knowledge-Base Poisoning and System-of-Record Validation in Utility Operations RAG

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Abstract—Retrieval-augmented generation (RAG) enables document-grounded technical lookup, but indexed repositories may accept low-trust contributions. We study targeted numeric answer-integrity corruption in utility-operations RAG with GridPoison-100: 35 source-grounded public items, 45 fictional asset-register items, and 20 legitimate-conflict items. The two pre-specified primary tests were null: fixed-context forced exposure produced 0% attacker-value adoption in both tiers, and a keyed system-of-record (SoR) check at 70% coverage did not significantly outperform a cautious prompt or isolate-then-aggregate voting (all Holm-adjusted $p = 1.0$). Descriptive natural-retrieval analysis shows a different operational effect. Under a base prompt that returns **VALUE: UNKNOWN** on unresolved conflict, one poison at $k = 5$ reached 70.0% of public-item queries; among exposed trials CAR was 4.1%, while Tier 1 CRR fell from 75.7–78.6% to 8.6–12.9%, primarily through non-commitment. In the pre-specified secondary $k = 2$ condition, CAR rose to 34.6% on the same 9 of 26 exposed trials for all three models. D3 is an upper-bound keyed record check: full coverage removed observed out-of-range commitments by construction, but flagged 64.1–65.7% of otherwise-correct legitimate-conflict answers because 13 of 20 conflict records held generic profiles. The results locate answer-integrity risk in retrieval composition and record scope, not model choice alone.

Index Terms—retrieval-augmented generation, knowledge-base poisoning, power systems, utility operations, large language models, data integrity

I. INTRODUCTION

Large language models are increasingly proposed as natural-language interfaces to technical documentation. In power and utility settings, the most plausible near-term use is not autonomous control but constrained lookup: a field worker asks for an approach distance, a protection engineer checks an interconnection boundary, or a planner retrieves an approved asset limit. RAG makes such answers auditable by grounding them in external documents, yet it also creates a write-path dependency. If an indexed repository accepts contributions from contractors, shared drives, wikis, or externally supplied bulletins, a legitimate user can receive an incorrect value even when the user’s prompt is benign.

Knowledge-base poisoning of RAG is already established [1], [2]. Accordingly, our contribution is not another demonstration that injected text can influence an answer. Instead, we measure how a conflict-aware utility assistant separates silent corruption from non-commitment, and how a separate structured record trades filtering benefit against false flags on legitimate jurisdiction- and interconnection-specific answers.

We study three research questions. **RQ1** asks how targeted poisoning decomposes into retrieval exposure and conditional adoption. **RQ2** asks whether attacker-value adoption differs between public-standard facts and asset-specific facts when exposure is held constant. **RQ3** asks how much system-of-record validation reduces out-of-range commitments relative to two generic defenses, and what false-flag cost it introduces when legitimate documents disagree by jurisdiction or inter-connection.

The contributions are fourfold: (1) a write-path threat model for low-trust utility-document repositories; (2) GridPoison-100, a benchmark with source-grounded public items, a fictional asset register, and legitimate-conflict controls; (3) a two-stage evaluation that separates retrieval compromise, conditional adoption, and consequence grading; and (4) an accounting of what deterministic system-of-record validation gains and costs under partial coverage and deliberately simplified record scope.

II. RELATED WORK

PoisonedRAG showed that a small number of attacker-controlled texts can steer retrieval-augmented models toward targeted answers [1]. Subsequent work has examined retrieval, database, and generator factors that influence poisoning exposure and attack success [2]. Blocker documents can also jam RAG systems into declining to answer [3]; the non-commitment observed here is related but arises from a conflict-abstention rule rather than an attacker objective. Because those effects are already known, retrieval depth is a secondary factor here rather than a novelty claim.

RobustRAG introduced an isolate-then-aggregate strategy for retrieval corruption [4]; we use this family as a generic defense baseline rather than presenting it as new. ClashEval quantified conflicts between parametric knowledge and retrieved evidence [5]. Other work has shown that evidence phrasing, publication time, and metadata can affect model preferences [6], [7]. These findings motivate our vendor-bulletin condition but also rule out treating authority framing as a primary contribution.

LLM risks in power systems have already been discussed and experimentally explored [8], [9]. Our study therefore avoids claims of being the first LLM-security work in the domain. The domain-specific contribution is the joint measurement of silent corruption, prompt-mediated non-

commitment, and record-scope false flags on source-grounded utility lookups.

III. THREAT MODEL

The attacker is an authorized but low-trust writer, such as a contractor with write access to an indexed collaboration repository, or the sender of a spoofed technical bulletin that is subsequently indexed. The attacker may add one to three new documents but cannot edit or delete existing documents. They cannot modify the structured asset/settings record, EMS/SCADA, protection devices, model weights, or the user’s query. The attacker knows the target asset or topic and can read existing documents on it, but does not know the exact query wording, embedding model, or retrieval results.

The attack goal is targeted answer-integrity corruption: a legitimate lookup should commit to an attacker-chosen numeric value. The primary poison is a factual statement rather than an instruction to the model, separating the main condition from indirect prompt injection. One stress paraphrase inherits the directive wording of the clean training-card template but changes only the value. Confidentiality, model-weight poisoning, multi-turn social engineering, and physical control-loop attacks are outside scope. Denial of service is not an attack goal, but we report loss of correct answers because it is the dominant observed effect.

IV. GRIDPOISON-100

GridPoison-100 contains 100 lookup items divided into three tiers. Tier 1 contains 35 public normative facts drawn from Massachusetts DER settings and return-to-service material, OSHA 29 CFR 1910.269 minimum-approach-distance tables, and NERC PRC-024-3 interconnection boundaries [10], [11], [12], [13]. Each public item records a source URL, source locator, units, applicability conditions, and an allowable interval.

Tier 2 contains 45 asset-specific facts from a declared fictional utility register. Asset identifiers and values are synthetic by construction, preventing a model from knowing them from pretraining. The register covers transformer emergency ratings, breaker current ratings, feeder thermal limits, BESS continuous discharge limits, and cable ampacity values. For Tier 2, the register is the authoritative record by construction; no claim is made that these values describe a real network.

Tier 3 contains 20 legitimate-conflict items. Two documents can disagree because of jurisdiction (Massachusetts versus IEEE 1547 defaults), NERC interconnection, or table scope (Quebec versus common PRC-024-3 tables), while the query explicitly names which context applies. These items are not poisoned; they measure whether a defense incorrectly flags a correct context-specific answer.

The clean corpus contains 400 short template-generated documents: three support documents per poisonable item in three styles (reference note, lookup summary, and training card), 40 legitimate-conflict documents, and 120 on-topic distractors; each distractor restates another item’s value under a different topic. Documents carry item and asset identifiers,

so the corpus is cleaner and more explicit than a real utility repository. Each poison is a minimal pair of one support document that changes only the critical value, making the conflict maximally visible. The main poison moves the value outside the encoded benchmark interval; a wrong-but-in-range variant separates factual incorrectness from interval violation, and a dated vendor-bulletin form tests authority framing.

V. EXPERIMENTAL DESIGN

A. Reference conditions and models

B0 is closed-book generation, B1 is clean RAG, and B2 is poisoned RAG with the same base prompt. The base prompt requests one committed numeric value when evidence is unambiguous and `VALUE: UNKNOWN` when supplied sources are inconsistent; B2 therefore already permits abstention on conflict, while D1 adds an explicit warning that retrieved sources may be falsified. Retrieval uses sentence-transformers/all-MiniLM-L6-v2 embeddings with cosine similarity and one document per chunk. The evaluated models are Qwen3-8B, Mistral Small 3.2 24B, and GPT-5.6 Luna, called through OpenRouter at temperature zero with `max_tokens=256` and provider-default reasoning settings. Provider token accounting did not enforce that request as a hard reasoning bound for Qwen, while GPT-5.6 Luna reasoning consumed the requested budget. Provider routing was not pinned; resolved model/provider metadata, response ID, token usage, and timestamp were logged per response.

B. Stage A: retrieval exposure

RQ1 uses natural dense retrieval. The lower-strength condition inserts one out-of-range poison; the stress condition inserts three distinct poison paraphrases. `Poison@k` is reported separately from generation behavior. A retrieval-only pre-generation calibration found that one-document exposure was strongly imbalanced between Tier 1 and Tier 2, which would confound a direct cross-tier adoption comparison. This exposure gap is a property of this corpus and retriever, not evidence that asset-specific facts resist poisoning.

C. Stage B: forced exposure

RQ2 therefore uses a fixed five-passage context rather than natural retrieval. Each item receives three clean same-item support passages, one deterministic on-topic distractor, and exactly one poison. The poison is counterbalanced in the first or last position. This deliberately evidence-rich design holds exposure and context length constant while comparing public-standard and asset-specific facts; it is not intended to represent natural attack strength.

D. Defenses

D1 is a cautious prompt instructing the model to identify unresolved source conflicts. D2 isolates each retrieved passage, obtains one value per passage with the cautious prompt, and returns the plurality value; ties yield `UNKNOWN`. It is a simplified instance of the isolate-then-aggregate family [4], not RobustRAG’s certified aggregation. D3 is deterministic

system-of-record validation: if the item’s record is covered and the committed value differs from it, the answer is replaced by a verification flag. D3 is given the applicable record key, so query-to-record linking errors are not modeled. For 78 of the 80 poisonable items the record equals the benchmark target; the exceptions are two Massachusetts undervoltage settings for which the simplified record stores IEEE 1547-2018 Category II defaults [14], [12]. For 13 of the 20 Tier 3 items the record stores a generic alternative rather than the applicable contextual variant. D3 therefore approximates an answer-key check on poisonable items; its purpose here is to quantify an upper-bound filtering benefit and the cost of record-scope mismatch, not to propose a new defense algorithm. Coverage at nominal 40%, 70%, and 100% is fixed deterministically by hashing item identifiers, realizing 30, 57, and 80 of 80 poisonable items and 6, 14, and 20 of 20 Tier 3 items.

RQ3 uses the three-poison $k = 5$ stress condition to avoid a retrieval floor when comparing defenses. The paraphrases rarely co-rank, so the stress condition mainly raises exposure rather than creating a poison majority. Secondary one-factor deviations examine a dated vendor bulletin, a wrong-but-in-range value, and $k = 2$. No full factorial sweep is performed.

VI. METRICS AND STATISTICAL ANALYSIS

The raw logs retain labels for correct (C), attacker-value adoption (A), another committed value (W), conflict flag (F), and refusal (R). For reported metrics, numeric C/A/W grading is deterministic from the required VALUE: line and item tolerance, while every response without a parsed numeric value is pooled as non-commitment (N); we do not interpret the parser-level F/R split.

Let E_t indicate whether a poisoned document occurs in the retrieved top- k context. We report Poison@ k , attack success rate (ASR), conditional adoption rate (CAR), correct-response rate (CRR), non-commitment rate (NCR), and benchmark-violation rate (BVR):

$$\text{CAR} = P(A \mid E_t = 1), \quad (1)$$

$$\text{BVR} = \frac{1}{|T|} \sum_t \mathbb{I}[\hat{v}_t \notin [L_i, U_i]]. \quad (2)$$

BVR counts only committed numeric values; non-committal outputs are non-violations by definition, so BVR is reported together with CRR. It is a benchmark consequence metric, not a physical grid-safety measure.

The item is the unit of analysis and the two query phrasings are repeated observations. Defense differences receive paired 95% item-cluster bootstrap intervals with 2,000 resamples; natural-retrieval CAR intervals use item-cluster resampling, and zero-event Stage-B adoption cells receive exact Clopper-Pearson bounds. The locked primary RQ3 comparison is D3 at 70% coverage versus D1 and D2. Paired comparisons use item-level exact McNemar tests, with an item counted as violating only when both phrasings violate, and Holm correction over the six primary comparisons; comparisons with B2 are secondary and descriptive. Tier 3 BFFR is conditioned

on clean-RAG correctness so D3 is charged only when it flags an otherwise-correct context-specific answer.

VII. RESULTS

Both pre-specified primary endpoints were null: Stage-B attacker-value adoption was 0% in both tiers (RQ2), and no D3@70 comparison with D1 or D2 was significant (RQ3). The exposed-trial breakdown of natural retrieval below was added after these outcomes were known and is descriptive; $k = 2$ was pre-specified as a secondary one-factor deviation. Clean-RAG CRR was 87.8% on Tier 2 (pre-specified threshold: 80%) and 75.7–78.6% on Tier 1.

A. Natural-retrieval behavior (RQ1)

Table I reports retrieval exposure and generation outcomes. Exposure is deterministic for the shared retriever; CRR and CAR are ranges across models. At $k = 5$, one poison was retrieved in 49 Tier 1 trials per model; only 0–3 remained correct, while direct attacker adoption was 4.1% (item-cluster 95% CI [0.0, 10.6]) of exposed trials. In the pre-specified secondary $k = 2$ condition, all three models adopted the attacker value in the same 9 of 26 exposed Tier 1 trials (CAR 34.6%; item-cluster 95% CI [17.9, 54.5]). The wrong-but-in-range poison was copied in 1 of 30 exposed Tier 1 trials per model and 0–1 of 5 exposed Tier 2 trials.

B. Fixed-context exposure (RQ2)

In the evidence-rich fixed context (three clean same-item supports, one distractor, one poison), attacker-value adoption and BVR were 0% in every model/tier/position cell. With zero events, the exact 95% upper bounds on per-item adoption are 10.0% for Tier 1 ($n = 35$) and 7.9% for Tier 2 ($n = 45$). Table II reports correct-answer retention. Every poison-last trial produced no numeric commitment. Because the base prompt explicitly permits VALUE: UNKNOWN when evidence is inconsistent, this is a conflict-handling result rather than evidence of inherent poisoning robustness.

C. Record-check accounting (RQ3)

The locked primary comparison is D3 at 70% coverage versus D1 and D2. Table III reports paired BVR reduction in percentage points; positive values favor D3. Against B2, which uses the same conflict-abstaining base prompt, D3@70 removed 6/7, 5/6, and 6/8 violating trials for Mistral, Qwen, and GPT-5.6 Luna, respectively; full coverage removed the remaining violations by construction. D2 increased Qwen BVR by 3.1 points relative to B2. Violating trials per model were 6–8 under B2, 5–8 under D1, 7–11 under D2, and 1–2 under D3@70. The primary McNemar comparisons had at most 3 discordant items and all six Holm-adjusted tests were non-significant ($p_{\text{Holm}} = 1.0$). The defense comparisons are descriptive. Counting an item as violating when either phrasing violates also leaves all six comparisons non-significant (minimum $p_{\text{Holm}} = 0.094$).

TABLE I

NATURAL-RETRIEVAL OUTCOMES (%; $k = 5$ UNLESS NOTED). CAR IS CONDITIONAL ON EXPOSURE; EXPOSED TIER 1/TIER 2 TRIALS PER MODEL ARE 49/3 (1 POISON), 49/69 (3 POISONS), 47/7 (BULLETIN), AND 26/3 ($k = 2$).

Condition	Exp.	Tier 1 (public)		Exp.	Tier 2 (asset)	
		CRR	CAR		CRR	CAR
Clean RAG	—	75.7–78.6%	—	—	87.8%	—
1 poison	70.0%	8.6–12.9%	4.1%	3.3%	84.4%	0.0%
3 poisons	70.0%	7.1–15.7%	4.1%	76.7%	14.4%	4.3–5.8%
1 bulletin	67.1%	20.0–22.9%	6.4–8.5%	7.8%	80.0%	0.0–14.3%
1 poison, $k = 2$	37.1%	14.3%	34.6%	3.3%	62.2%	0.0%

TABLE II
CORRECT-ANSWER RATE UNDER FIXED EXPOSURE (%).

Model	First/T1	First/T2	Last/T1	Last/T2
Mistral Small 3.2	8.6%	97.8%	0.0%	0.0%
Qwen3-8B	42.9%	57.8%	0.0%	0.0%
GPT-5.6 Luna	74.3%	28.9%	0.0%	0.0%

D. Legitimate-conflict cost

Clean RAG answered 35–39 of 40 legitimate-conflict trials correctly. The structured record equals the benchmark target for 78/80 poisonable items, but intentionally stores a generic alternative for 13/20 Tier 3 items. BFFR therefore measures a controlled record-scope gap and is conditioned on clean-RAG correctness. At 70% coverage, conditional BFFR was 43.6–45.7%; at full coverage it was 64.1–65.7%. The cost is therefore driven by record scope, not only by model behavior.

E. Prospective Follow-up and Exploratory Backup

After freezing the original Qwen/Mistral/Luna primary results, we locked additional prompt, retriever, and context-composition controls. The Qwen follow-up did not finish because of repeated API failures. We retained the completed Mistral and Luna controls and added Gemini 2.5 Flash-Lite as an exploratory emergency replacement, using its own matched B2 baseline. Gemini was not a pre-specified model family and is excluded from all original confirmatory results.

The dense rerun reproduced 1920 historical generation rows with 0 poison-rank mismatches. When the prompt required a numeric commitment, the paired exposed-trial results were Mistral Small 3.2: 49.0% CAR ($p = 4.77 \times 10^{-7}$; material prompt mediation supported); GPT-5.6 Luna: 16.3% CAR ($p = 0.0312$; partial prompt mediation); Gemini 2.5 Flash-Lite (expl.): 42.9% CAR ($p = 4.01 \times 10^{-5}$; material prompt mediation supported). At least one completed follow-up model failed the pre-stated composition thresholds: C1/C2 adoption reached 20% or C4 did not exceed 50% adoption. Gemini is exploratory and does not replace Qwen in the locked model family; all three observed C1/C2/C4 rates are shown. BM25 is reported as a retriever sensitivity rather than a direct robustness claim because its poison-exposure rate differs from dense retrieval; CAR is therefore interpreted together with Poison@ k . A separate post-hoc benchmark-consequence mapping classified 18 OSHA MAD poisons: 7 fell below the benchmarked

inadvertent-movement allowance and 11 consumed part of the electrical-distance component while remaining at or above that allowance. All 7 PRC-024-3 poisons shortened the benchmark no-trip time by 60%. All 45 fictional Tier-2 poisons exceeded the record limit by 20%. These are benchmark consequence classes, not professional-engineering or physical-safety determinations. The post-hoc severity-prioritized D3 what-if did not improve coverage monotonically. Relative to the original hash selection at matched realized coverage, BVR changed by Qwen3-8B -0.625 pp, Mistral Small 3.2 -0.625 pp, GPT-5.6 Luna +0 pp at 40%, and Qwen3-8B +0.625 pp, Mistral Small 3.2 +1.25 pp, GPT-5.6 Luna +0.625 pp at 70%. Tier-3 hash coverage was held fixed, so this comparison changes only which poisonable records are covered.

VIII. DISCUSSION

Natural retrieval shows how poisoning manifests under a conflict-abstaining prompt. At $k = 5$, a retrieved poison rarely transferred its value, but exposed Tier 1 trials usually lost the correct answer and became non-committal. Operationally these are different failures: non-commitment is an availability loss that routes the lookup to human or record verification, whereas attacker-value adoption is a silent integrity failure. In the pre-specified secondary $k = 2$ condition, all three model families adopted the attacker value on the same nine exposed trials, consistent with a retrieval-composition effect rather than a model-specific one. Because retrieved document IDs were not logged in generation rows, we do not claim that a particular clean-copy configuration caused those nine adoptions.

RQ2 was inconclusive rather than evidence of robustness: forced-exposure adoption was at floor in both tiers. The evidence-rich Stage-B context contains three clean same-item supports against one poison, and the base prompt explicitly permits abstention on unresolved conflict. Closed-book Tier 1 CRR was 0–2.9%, but GPT-5.6 Luna returned empty output in 41 of 70 closed-book Tier 1 trials under the requested token budget, so its parametric prior was not adequately measured.

D3 measures record coverage rather than language-model behavior. Because it is given the applicable record key and the record equals the target for 78/80 poisonable items, full coverage removes observed violations by construction. Under independent random coverage c , expected residual violations scale with the uncovered fraction $(1 - c)$; our deterministic hash is one realized coverage draw, with 57/80 poisonable items covered at the nominal 70% level. The locked D3-vs-

TABLE III

PRIMARY RQ3 COMPARISON AND VIOLATION COUNTS. BRACKETS ARE DESCRIPTIVE 95% ITEM-CLUSTER BOOTSTRAP INTERVALS OF TRIAL-LEVEL BVR DIFFERENCES; MCNEMAR TESTS COUNT AN ITEM AS VIOLATING ONLY WHEN BOTH PHRASINGS VIOLATE.

Model	D3@70 vs D1 (pp)	D3@70 vs D2 (pp)	B2 → D3@70	D3@100
Mistral Small 3.2	2.5 [-0.6, 6.2]	3.8 [0.6, 8.1]	7 → 1	0
Qwen3-8B	3.1 [0.6, 6.9]	6.2 [1.9, 11.2]	6 → 1	0
GPT-5.6 Luna	3.8 [0.6, 7.5]	3.8 [0.0, 8.8]	8 → 2	0

TABLE IV

CONDITIONAL BENIGN-CONFLICT FALSE-FLAG RATE (BFFR).

Model	40%	70%	100%
Mistral Small 3.2	17.9%	43.6%	64.1%
Qwen3-8B	17.1%	45.7%	65.7%
GPT-5.6 Luna	17.9%	43.6%	64.1%

TABLE V

FOLLOW-UP CONTROLS (PERCENT). GEMINI IS AN EXPLORATORY BACKUP; BM25 P@2/CAR DENOTES EXPOSURE/CONDITIONAL ADOPTION.

Model	Commit CAR	BM25 P@2/CAR	C1	C2	C4
Mistral Small 3.2	49.0	1.2/0.0	0.0	21.2	100.0
GPT-5.6 Luna	16.3	1.2/0.0	0.0	18.8	100.0
Gemini 2.5 Flash-Lite (expl.)	42.9	1.2/0.0	1.2	78.8	100.0

D1/D2 comparisons involve few violating items and remain non-significant, so they are descriptive.

The Tier 3 control gives the more operationally useful record-scope result. Clean RAG answered 35–39 of 40 scoped-conflict trials correctly, showing that the models usually resolved legitimate disagreements when the query named the applicable jurisdiction or interconnection. A generic SoR then flags many of those correct answers: 13 of 20 Tier 3 records intentionally omit the applicable contextual variant. A structured check therefore helps only when record authority, jurisdiction/interconnection scope, and escalation policy are explicit.

IX. LIMITATIONS

GridPoison-100 is a controlled template benchmark, not a deployed utility corpus: documents carry explicit identifiers, minimal-pair poisons make conflicts visible, and the base prompt sanctions abstention. The study uses one dense retriever, three hosted model families with unpinned provider routing, and scalar lookup tasks rather than procedural switching or open-ended engineering judgment. GPT-5.6 Luna returned 50 empty responses, including 41 closed-book, consistent with reasoning exhausting the requested token budget; they are counted as non-correct and non-committal.

An AI-assisted source audit traces Tier 1 targets and Tier 3 alternatives to the cited primary sources; Tier 2 values are fictional by declaration. PRC-024-3 is used as the cited benchmark edition, not asserted as current operational guidance. The encoded intervals are benchmark definitions, not engineering operating limits, and no power-flow, protection, thermal, or

transient simulation is performed. D3 is an upper bound because the harness supplies the applicable record key.

X. CONCLUSION

GridPoison-100 separates retrieval exposure, attacker-value adoption, non-commitment, and benchmark consequence in utility-operations RAG. Under a conflict-abstaining prompt, one retrieved poison at $k = 5$ usually converted a correct lookup into non-commitment rather than silent corruption; in the pre-specified secondary $k = 2$ condition, adoption rose and occurred on the same nine exposed trials across all three model families. A keyed structured-record check removed covered out-of-range commitments in proportion to its coverage, but flagged correct answers whenever the record lacked the applicable jurisdiction or interconnection variant. For utility deployments, the practical implication is architectural: preserve redundant approved evidence in retrieval, cover high-consequence settings in the structured record, and scope those records to the jurisdiction and interconnection in which they are authoritative.

GENERATIVE AI USE DISCLOSURE

Generative AI systems including ChatGPT (OpenAI), Claude (Anthropic), and Grok (xAI) assisted software scaffolding, benchmark/corpus drafting, literature triage, source-consistency checking, adversarial methodological review, and language editing. All reported numerical results derive from logged experiments and deterministic analysis code; the authors remain fully responsible for the accuracy, originality, and integrity of the manuscript.

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